

5.5 For each of the following two time-series regression models, and assuming MR1–MR6 hold, find $\text{var}(b_2|\mathbf{x})$ and examine whether the least squares estimator is consistent by checking whether $\lim_{T \rightarrow \infty} \text{var}(b_2|\mathbf{x}) = 0$.

- a. $y_t = \beta_1 + \beta_2 t + e_t, t = 1, 2, \dots, T$. Note that $\mathbf{x} = (1, 2, \dots, T)$, $\sum_{t=1}^T (t - \bar{t})^2 = \sum_{t=1}^T t^2 - \left(\sum_{t=1}^T t\right)^2 / T$, $\sum_{t=1}^T t = T(T+1)/2$ and $\sum_{t=1}^T t^2 = T(T+1)(2T+1)/6$.
- b. $y_t = \beta_1 + \beta_2 (0.5)^t + e_t, t = 1, 2, \dots, T$. Here, $\mathbf{x} = (0.5, 0.5^2, \dots, 0.5^T)$. Note that the sum of a geometric progression with first term r and common ratio r is

$$\text{等比級數: } S = r + r^2 + r^3 + \dots + r^n = \frac{r(1-r^n)}{1-r}$$

- c. Provide an intuitive explanation for these results.

$$(a) \sum_{t=1}^T (t - \bar{t})^2 = \sum t^2 - \frac{(\sum t)^2}{T} \quad \text{又 } \sum t = \frac{T(T+1)}{2}, \quad \sum t^2 = \frac{T(T+1)(2T+1)}{6}$$

$$\Rightarrow \sum (t - \bar{t})^2 = \frac{T(T+1)(2T+1)}{6} - \frac{\left(\frac{T(T+1)}{2}\right)^2}{T} = \frac{T(T^2-1)}{12}$$

$$\text{Var}(b_2|\mathbf{x}) = \frac{\sigma^2}{\sum (t - \bar{t})^2} = \frac{12\sigma^2}{T(T^2-1)}$$

As $T \rightarrow \infty$, $\text{var}(b_2|\mathbf{x}) \rightarrow 0$, and thus the least squares estimator is consistent.

$$(b) \sum_{t=1}^T x_t = \sum (0.5)^t = \frac{0.5(1-(0.5)^T)}{1-0.5} = 1-(0.5)^T \Rightarrow \text{As } T \rightarrow \infty, \sum x_t \rightarrow 1$$

$$\sum x_t^2 = \sum (0.5)^{2t} = \sum (0.25)^t = \frac{0.25(1-(0.25)^T)}{1-0.25} \Rightarrow \frac{0.25}{0.75} = \frac{1}{3}$$

$$\text{var}(b_2|\mathbf{x}) = \frac{\sigma^2}{\frac{1}{3}(1-(0.5)^{2T}) - \frac{(1-(0.5)^T)^2}{T}}$$

As $T \rightarrow \infty$, $\text{var}(b_2|\mathbf{x}) \rightarrow 3\sigma^2$. Since $\lim_{T \rightarrow \infty} \text{var}(b_2|\mathbf{x}) \neq 0$, the least squares estimator b_2 is not consistent.

- (c) In part (a), increasing T provides more and more information about β_2 and the relationship between y and x . However, in part (b), $x_t = 0.5^t$ becomes zero for moderately sized T , and increasing T further does not provide any more information about β_2 .

$$n=63, df=63-3=60, \alpha=5\%, t_{(0.025, 60)} \doteq 2$$

5.6 Suppose that, from a sample of 63 observations, the least squares estimates and the corresponding estimated covariance matrix are given by

$$\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix} \quad \widehat{\text{cov}}(b_1, b_2, b_3) = \begin{bmatrix} 3 & -2 & 1 \\ -2 & 4 & 0 \\ 1 & 0 & 3 \end{bmatrix}$$

Using a 5% significance level, and an alternative hypothesis that the equality does not hold, test each of the following null hypotheses:

- a. $\beta_2 = 0$
- b. $\beta_1 + 2\beta_2 = 5$
- c. $\beta_1 - \beta_2 + \beta_3 = 4$

(a)
$$\begin{cases} H_0: \beta_2 = 0 \\ H_1: \beta_2 \neq 0 \end{cases}$$

$$\text{Var}(\hat{\beta}_2) = 4 \rightarrow \text{se}(\beta_2) = 2$$

$$t = \frac{3-0}{2} = 1.5, \quad 1.5 < 2 \rightarrow \text{fail to reject } H_0$$

(b)

$$\begin{cases} H_0: \beta_1 + 2\beta_2 = 5 \\ H_1: \beta_1 + 2\beta_2 \neq 5 \end{cases}$$

估計值: $2 + 2 \times 3 = 8$

$$\begin{aligned} \text{Var}(b_1 + 2b_2) &= \text{Var}(b_1) + 4\text{Var}(b_2) + 4\text{cov}(b_1, b_2) \\ &= 3 + 4 \times 4 + 4 \times (-2) = 11 \end{aligned}$$

$$\text{se}(b_1 + 2b_2) = \sqrt{11} \doteq 3.317$$

$$t = \frac{8-5}{3.317} \doteq 0.904, \quad 0.904 < 2 \rightarrow \text{fail to reject } H_0$$

(c)

$$\begin{cases} H_0: \beta_1 - \beta_2 + \beta_3 = 4 \\ H_1: \beta_1 - \beta_2 + \beta_3 \neq 4 \end{cases}$$

估計值: $2 - 3 + (-1) = -2$

$$\begin{aligned} \text{Var}(b_1 - b_2 + b_3) &= \text{Var}(b_1) + \text{Var}(b_2) + \text{Var}(b_3) - 2\text{cov}(b_1, b_2) \\ &\quad + 2\text{cov}(b_1, b_3) - 2\text{cov}(b_2, b_3) \\ &= 3 + 4 + 3 - 2 \times (-2) + 2 - 2 \times 0 = 16 \end{aligned}$$

$$\text{se}(b_1 - b_2 + b_3) = \sqrt{16} = 4$$

$$t = \frac{-2-4}{4} = -1.5, \quad |-1.5| < 2 \rightarrow \text{fail to reject } H_0$$

5.20 A generalized version of the estimator for β_2 proposed in Exercise 2.9 by Professor I.M. Mean for the regression model $y_i = \beta_1 + \beta_2 x_i + e_i, i = 1, 2, \dots, N$ is

$$\hat{\beta}_{2,mean} = \frac{\bar{y}_2 - \bar{y}_1}{\bar{x}_2 - \bar{x}_1}$$

where $(\bar{y}_1, \bar{x}_1)$ and $(\bar{y}_2, \bar{x}_2)$ are the sample means for the first and second halves of the sample observations, respectively, after ordering the observations according to increasing values of x . Given that assumptions MR1–MR6 hold:

- Show that $\hat{\beta}_{2,mean}$ is unbiased.
- Derive an expression for $\text{var}(\hat{\beta}_{2,mean} | \mathbf{x})$.
- Write down an expression for $\text{var}(\hat{\beta}_{2,mean})$.
- Under what conditions will $\hat{\beta}_{2,mean}$ be a consistent estimator for β_2 ?
- Randomly generate observations on x from a uniform distribution on the interval $(0, 10)$ for sample sizes $N = 100, 500, 1000$, and, if your software permits, $N = 5000$. Assuming $\sigma^2 = 1000$, for each sample size, compute:
 - $\text{var}(b_2 | \mathbf{x})$ and $\text{var}(\hat{\beta}_{2,mean} | \mathbf{x})$ where b_2 is the OLS estimator.
 - Estimates for $E[(s_x^2)^{-1}]$ and $E[4/(\bar{x}_2 - \bar{x}_1)^2]$ where s_x^2 is the sample standard deviation for x using N as a divisor.
- Comment on the relative magnitudes of your answers in part (e), (i) and (ii) and how they change as sample size increases. Does it appear that $\hat{\beta}_{2,mean}$ is consistent?
- Show that $E[(s_x^2)^{-1}] \xrightarrow{p} 0.12$ and $E[4/(\bar{x}_2 - \bar{x}_1)^2] \xrightarrow{p} 0.16$. [Hint: The variance of a uniform random variable defined on the interval (a, b) is $(b - a)^2/12$.]
- Suppose that the observations on x were not ordered according to increasing magnitude but were randomly assigned to any position. Would the estimator $\hat{\beta}_{2,mean}$ be consistent? Why or why not?

(a)

$$y_i = \beta_1 + \beta_2 x_i + e_i, \quad E(e_i | X) = 0$$

$$\bar{y}_1 = \beta_1 + \beta_2 \bar{x}_1 + \bar{e}_1, \quad \bar{y}_2 = \beta_1 + \beta_2 \bar{x}_2 + \bar{e}_2$$

$$\bar{y}_2 - \bar{y}_1 = \beta_2 (\bar{x}_2 - \bar{x}_1) + (\bar{e}_2 - \bar{e}_1)$$

$$\Rightarrow \hat{\beta}_{2,mean} = \frac{\bar{y}_2 - \bar{y}_1}{\bar{x}_2 - \bar{x}_1} = \beta_2 + \frac{\bar{e}_2 - \bar{e}_1}{\bar{x}_2 - \bar{x}_1}$$

$$E[\hat{\beta}_{2,mean} | X] = \beta_2 + \frac{E[\bar{e}_2 | X] - E[\bar{e}_1 | X]}{\bar{x}_2 - \bar{x}_1} = \beta_2 + 0 = \beta_2$$

$$\therefore E[\hat{\beta}_{2,mean} | X] = \beta_2 \rightarrow \text{unbiased} *$$

(b)

$$\text{Var}(\hat{\beta}_{2, \text{mean}} | X) = \frac{\text{Var}(\bar{e}_2 - \bar{e}_1)}{(\bar{x}_2 - \bar{x}_1)^2}, \quad e_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$$

$$\text{Var}(\bar{e}_1 | X) = \frac{\sigma^2}{(\frac{N}{2})} = \frac{2\sigma^2}{N}, \quad \text{Var}(\bar{e}_2 | X) = \frac{\sigma^2}{(\frac{N}{2})} = \frac{2\sigma^2}{N}$$

$$\Rightarrow \text{Var}(\bar{e}_2 - \bar{e}_1 | X) = \frac{2\sigma^2}{N} + \frac{2\sigma^2}{N} = \frac{4\sigma^2}{N} \quad (\because 2 \text{ 組 } e \text{ 各自獨立})$$

$$\therefore \text{Var}(\hat{\beta}_{2, \text{mean}} | X) = \frac{\text{Var}(\bar{e}_2 - \bar{e}_1 | X)}{(\bar{x}_2 - \bar{x}_1)^2} = \frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2} \quad *$$

(c)

$$\text{Var}(\hat{\beta}_{2, \text{mean}}) = E[\text{Var}(\hat{\beta}_{2, \text{mean}} | X) + \text{Var}(E[\hat{\beta}_{2, \text{mean}} | X])]$$

Also, $E[\hat{\beta}_{2, \text{mean}} | X] = \beta_2$ is constant and its variance is 0.

$$\text{Therefore, } \text{Var}(\hat{\beta}_{2, \text{mean}}) = E\left[\frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2}\right] \quad *$$

(d)

consistent : $N \rightarrow \infty$, variance $\rightarrow 0$

$$\Rightarrow \text{Var}(\hat{\beta}_{2, \text{mean}} | X) \rightarrow 0$$

$$\frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2} \rightarrow 0 \quad \Rightarrow (\bar{x}_2 - \bar{x}_1) \text{ 不趨近 } 0. \quad *$$

(e)

```
> print(results)
  N   Var_OLS Var_Mean Inv_Sx2 Four_Diff2
1 100 1.24362517 1.6631963 0.1243625 0.1663196
2 500 0.24402399 0.3260404 0.1220120 0.1630202
3 1000 0.12290628 0.1656282 0.1229063 0.1656282
4 5000 0.02413174 0.0323161 0.1206587 0.1615805
>
> # 額外計算：相對效率 (OLS 永遠比較有效)
> results$Efficiency_Ratio <- results$Var_OLS / results$Var_Mean
> print("OLS 與 Mean 估計式的效率比:")
[1] "OLS 與 Mean 估計式的效率比:"
> print(results[, c("N", "Efficiency_Ratio")])
  N Efficiency_Ratio
1 100      0.7477321
2 500      0.7484470
3 1000     0.7420615
4 5000     0.7467405
```

(i)

For every sample size ($N = 100, 500, 1000, 5000$), the OLS variance is consistently smaller than the Mean estimator variance

This aligns with the Gauss-Markov Theorem, that is the OLS estimator is the Best Linear Unbiased Estimator, meaning it has the minimum variance among all linear unbiased estimators.

(ii)

$E[(S_x^*)^{-1}]$: The value remains stable around 0.12.

$E[4/(\bar{x}_2 - \bar{x}_1)^2]$: The value remains stable around 0.16.

(f)

The simulation confirms that the variance of $\hat{\beta}_{2, \text{mean}}$ is approximately 1.33 times larger than that of the OLS estimator (since $0.16 / 0.12 \approx 4/3$).

As N increases from 100 to 5000, both Var_OLS and Var_Mean decrease significantly toward zero. Inv_Sx2 and Four_Diff2 converge to the theoretical limits of 0.12 and 0.16, respectively.

Yes, $\hat{\beta}_{2, \text{mean}}$ appears to be a consistent estimator.

Because the denominator $(\bar{x}_2 - \bar{x}_1)^2$ stays relatively constant while N in the denominator of the variance formula increases, the total variance will eventually reach zero in the limit.

(g)

As the sample size $N \rightarrow \infty$, we apply the law of large numbers and the properties of the Uniform distribution $U(a,b)$

$$\textcircled{1} E[(S_x^2)^{-1}] \xrightarrow{P} 0.12$$

$$\text{For } x \sim U(0,10), \sigma_x^2 = \frac{(b-a)^2}{12} = \frac{(10-0)^2}{12} = \frac{100}{12} \approx 8.333$$

By LLN, the sample variance S_x^2 converges in probability to the population variance, i.e. $S_x^2 \xrightarrow{P} 8.333$

$$\therefore (S_x^2)^{-1} \xrightarrow{P} \frac{1}{8.333} \approx 0.12$$

$$\textcircled{2} E\left[\frac{4}{(\bar{x}_2 - \bar{x}_1)^2}\right] \xrightarrow{P} 0.16$$

When observations are ordered, $\bar{x}_1$ represents the mean of the lower half of the $U(0,10)$ distribution, which converges to the expected value of the interval $[0,5]$, i.e. $E[X|X < 5] = 2.5$

Similarly, $\bar{x}_2$ converges to the expected value of the upper half interval $[5,10]$, i.e. $E[X|X > 5] = 7.5$

$$\therefore (\bar{x}_2 - \bar{x}_1)^2 \xrightarrow{P} (7.5 - 2.5)^2 = 25.$$

$$\frac{4}{(\bar{x}_2 - \bar{x}_1)^2} \xrightarrow{P} \frac{4}{25} = 0.16$$

(h) The estimator $\hat{\beta}_{2,\text{mean}}$ is not consistent.

$\therefore$ Without ordering, both $\bar{x}_1$ and $\bar{x}_2$ are random draws from the same population. As $N \rightarrow \infty$, $\bar{x}_1$ and $\bar{x}_2$ converges to the same population mean $E[X] = 5$, then $(\bar{x}_2 - \bar{x}_1) \rightarrow 0$.

$\text{Var}(\hat{\beta}_{2,\text{mean}} | X) = \frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2} \approx \frac{1}{N}$, fail to collapse onto the true parameter β_2 .